

Coherent growth of high-Miller-index facets enhances perovskite solar cells

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Shunde Li^{1,17}, Yun Xiao^{2,17}, Rui Su^{1,17}, Weidong Xu^{3,4,17}, Deying Luo⁵, Pengru Huang⁶, Linjie Dai³, Peng Chen¹, Pietro Caprioglio², Karim A. Elmestekawy², Milos Dubajic⁴, Cullen Chosy^{3,4}, Juntao Hu⁷, Irfan Habib², Akash Dasgupta², Dengyang Guo³, Yorrick Boeije^{3,4}, Szymon J. Zelewski^{3,4,8}, Zhangyuchang Lu¹, Tianyu Huang¹, Qiuyang Li¹, Jingmin Wang⁹, Haoming Yan¹, Hao-Hsin Chen¹, Chunsheng Li¹⁰, Barnaby A. I. Lewis^{3,4}, Dengke Wang⁷, Jiang Wu¹⁰, Lichen Zhao¹, Bing Han¹¹, Jianpu Wang^{9,12}, Laura M. Herz², James R. Durrant^{13,14}, Kostya S. Novoselov⁶, Zheng-Hong Lu^{7,15}, Qihuang Gong^{1,10,16}, Samuel D. Stranks^{3,4}, Henry J. Snaith² & Rui Zhu^{1,10,16}

Obtaining micron-thick perovskite films of high quality is key to realizing efficient and stable positive (p)-intrinsic (i)-negative (n) perovskite solar cells^{1,2}, but it remains a challenge. Here we report an effective method for producing high-quality, micron-thick formamidinium-based perovskite films by forming coherent grain boundaries, in which high-Miller-index-oriented grains grow on the low-Miller-index-oriented grains in a stabilized atmosphere. The resulting micron-thick perovskite films, with enhanced grain boundaries and grains, showed stable material properties and outstanding optoelectronic performances. The small-area solar cells achieved efficiencies of 26.1%. The 1-cm² devices and 5 cm × 5 cm mini-modules delivered efficiencies of 24.3% and 21.4%, respectively. The devices processed in a stabilized atmosphere presented a high reproducibility across all four seasons. The encapsulated devices exhibited superior long-term stability under both light and thermal stressors in ambient air.

Perovskite solar cells (PSCs) with a p-i-n configuration are one of the most promising clean-energy-harvesting photovoltaic technologies, owing to their low cost, facile fabrication and compatibility with monolithic tandem cells^{3–5}. Constructing the PSC with the p-type hole extraction layer first, followed by the intrinsic (i) perovskite absorber layer and then the n-type electron extraction layer, enables advantages when integrated into tandem solar cells and enables superior long-term stability, in comparison with n-i-p PSCs. However, the most recently certified power conversion efficiency (PCE) for p-i-n PSCs remains well below the detailed balance thermodynamic limit⁶. The PCE is the product of a short-circuit current density (J_{sc}), an open-circuit voltage (V_{oc}) and a fill factor (FF). Although the product $V_{oc} \times FF$ has reached 87.9% of the detailed balance thermodynamic limit⁷, the J_{sc} of p-i-n PSCs is below the highest values reported in the n-i-p counterpart^{8,9} (Supplementary Fig. 1 and Table 1). The relatively low J_{sc} for the p-i-n PSCs is partially a result of the thinner perovskite absorber usually employed (less than 500 nm) compared to the n-i-p PSCs with a thickness of over 700 nm¹⁰. Combined with low conduction charge-carrier

extraction layers usually employed in p-i-n PSCs, this makes it difficult to modulate the cavity interference inside the perovskite by increasing the thickness of charge-carrier extraction layers (used as optical spacers), thus lowering the J_{sc} . Although coating an anti-reflection layer on textured semi-transparent electrodes can reduce the optical reflection losses at the interfaces¹¹, it does not address the optical losses inside the perovskite absorber layer, especially in the long-wavelength range.

Beyond challenges in optical losses inside a thinner perovskite absorber layer, mitigating the degradation rate of perovskite absorber layers when operating in real-world working conditions is necessary to advance this technology towards commercialization. Despite notable improvements in the perovskite crystalline quality^{12,13} and defect passivation^{14–16}, alongside the selection and modification of charge-carrier extraction layers^{17,18}, the reported operational lifespan of p-i-n PSCs remains an order of magnitude lower than the required product lifetime of at least 25 years (ref. 19). The limited lifespan of p-i-n PSCs under operational conditions was thought to correlate with the intrinsic instability of perovskite absorber layers apart from charge-carrier

¹State Key Laboratory for Artificial Microstructure and Mesoscopic Physics, School of Physics, Frontiers Science Center for Nano-optoelectronics and Collaborative Innovation Center of Quantum Matter, Peking University, Beijing, China. ²Clarendon Laboratory, Department of Physics, University of Oxford, Oxford, UK. ³Cavendish Laboratory, University of Cambridge, Cambridge, UK. ⁴Department of Chemical Engineering and Biotechnology, University of Cambridge, Cambridge, UK. ⁵International Institute for Interdisciplinary and Frontiers, Beihang University, Beijing, China. ⁶Institute for Functional Intelligent Materials, National University of Singapore, Singapore, Singapore. ⁷Department of Physics, Center for Optoelectronics Engineering Research, Yunnan University, Kunming, China. ⁸Department of Experimental Physics, Faculty of Fundamental Problems of Technology, Wrocław University of Science and Technology, Wrocław, Poland. ⁹Key Laboratory of Flexible Electronics (KLOFE), Institute of Advanced Materials (IAM) and School of Flexible Electronics (Future Technologies), Nanjing Tech University (NanjingTech), Nanjing, China. ¹⁰Yangtze Delta Institute of Optoelectronics, Peking University, Nantong, China. ¹¹Eastern Institute for Advanced Study, Eastern Institute of Technology, Ningbo, China. ¹²School of Materials Science and Engineering and School of Microelectronics and Control Engineering, Changzhou University, Changzhou, China. ¹³Department of Chemistry and Centre for Processable Electronics, Imperial College London, London, UK. ¹⁴Department of Materials Science and Engineering, University of Swansea, Swansea, UK. ¹⁵Department of Materials Science and Engineering, University of Toronto, Toronto, Canada. ¹⁶Collaborative Innovation Center of Extreme Optics, Shanxi University, Taiyuan, China. ¹⁷These authors contributed equally: Shunde Li, Yun Xiao, Rui Su, Weidong Xu. [✉]e-mail: bhlady@buaa.edu.cn; bhan@eitech.edu.cn; qhgong@pku.edu.cn; sds65@cam.ac.uk; henry.snaith@physics.ox.ac.uk; iamzhurui@pku.edu.cn

extraction layers²⁰. Recently, evidence suggests that the initial degradation of perovskite absorber layers is influenced by the specific Miller-index facets exposed to external stressors²¹. The closely stacked Miller-index facets of formamidinium-based perovskite films, which are a compelling absorber for single-junction p-i-n PSCs, have proven to slow down the degradation rate once exposed to stressors such as moisture²¹. Compared with low-Miller-index facets, high-Miller-index facets with closely stacked atomic arrangements were shown to reduce imperfections with increasing surface terminations^{22,23}. The change in the oriented nucleation has been shown to influence the Miller-index orientation of the resultant perovskite absorber layer²⁴. We must recognize that the initial nucleation of formamidinium-based perovskite absorber layers correlates with the removal of the residual solvents, whereas the solvent evaporation rate is substantially influenced by the atmospheric processing temperature before heating samples on a hotplate²⁵. Therefore, engineering high-Miller-index orientations in state-of-the-art micron-thick perovskite films by optimizing whole processing conditions is a strategic move to advance this emerging photovoltaic technology.

Micron-thick perovskite films

Using detailed optical constants of the layers used in our full device stacks (Supplementary Fig. 2), we have simulated the light absorption profile of the devices (Supplementary Figs. 3 and 4 and Supplementary Note 1) and presented the calculated short-circuit current density (J_{sc}) as a function of the formamidinium-based perovskite film thickness in Fig. 1a. In our optical simulations, we used formamidinium-based perovskite composition as a base owing to their impressive device performance. Our optical simulations indicate that there is considerable scope for boosting the J_{sc} in the p-i-n PSC stacks by increasing the film thickness from 500 nm to 1,000 nm, with specific peaks at the film thicknesses of 600 nm, 800 nm and 1,000 nm. Yet, increased defects in micron-thick perovskite films may negate the potential optical benefit of increasing film thickness²⁶. Therefore, enhancing the quality of the micron-thick perovskite film is necessary.

We fabricated a series of devices featuring formamidinium-based perovskite films with thicknesses ranging from 200 nm to 1,100 nm. This was achieved by varying the perovskite precursor concentration using anti-solvent approaches, as detailed in Supplementary Table 2. Furthermore, controlling the atmospheric processing temperature presents another method to adjust the optimal film thickness. Hence, the combined effects of varying the precursor concentration and the atmospheric processing temperature would be particularly effective in making high-quality micron-thick perovskite films. The corresponding J_{sc} values alongside the simulated data are shown in Fig. 1a. The detailed device performance of making the PSCs is shown in Supplementary Fig. 5. Notably, the measured J_{sc} values do not represent 100% internal quantum efficiencies (IQEs), but there seems to be a region with film thicknesses between 600 nm and 1,000 nm in which the IQE of the devices is approximated to be around 93% but drops off when perovskite films are either thinner or thicker. Furthermore, our optical simulations suggest that the external quantum efficiency (EQE) spectrum for the device with the perovskite thickness of 1,000 nm exhibits long-wavelength enhancements (500–800 nm) in comparison with the previously typical thicknesses of about 500 nm, as shown in Fig. 1b. The long-wavelength enhancements in a wide range of 500–800 nm are because of improved optical cavity effects and enhanced optical power densities inside the perovskite layer with increasing thickness, which we show in Supplementary Figs. 3 and 4. Consistent light absorption spectra show an increase in a wide range of long wavelengths (even at the band edge of interest) when the perovskite thickness increases from 200 nm to 1,100 nm by varying the precursor concentration (Fig. 1c and Supplementary Fig. 7). The specific thickness of the formamidinium-based perovskite sample is controlled by

the synergistic effect of both the precursor concentration and the atmospheric processing temperature (Fig. 1d,e). Cross-sectional and top-view scanning electron microscope (SEM) images show that the micron-thick formamidinium-based perovskite films are pinhole-free both parallel and perpendicular to the substrates (Fig. 1d,e and Supplementary Fig. 8), even when using a higher perovskite concentration solution reaching 2.0 mol l⁻¹ (M) in a stabilized atmosphere (20–26 °C). This is in comparison with the previously typical perovskite concentration of 1.5 M or less in an ambient temperature^{7,27,28}. Notably, most grains are throughout the thickness of the micron-thick perovskite films, which would inhibit charge-carrier scattering and non-radiative recombination during the transport process²⁹. Details about materials and optoelectronic properties will be discussed below.

Engineering high-Miller-index facets

We made a series of micron-thick formamidinium-based perovskite samples in a stabilized atmosphere (Supplementary Fig. 9), and we characterized the material and optoelectronic properties of these samples. X-ray diffraction (XRD) data in Fig. 2a show that the formamidinium-based perovskite (without surface passivation) processed at a relatively low temperature (less than 26 °C) exhibits a high (211) peak at 34.9° and a low (001) peak at 14.1°, respectively. By contrast, when processed at a relatively high ambient temperature (26–29 °C) that is commonly presented in the device fabrication, the peak (001) facet is dominated in the XRD pattern. Combined with time-resolved photoluminescence (TRPL) decays (Supplementary Note 2), photoluminescence quantum yields (PLQYs) and the Urbach energy (E_U) determined by photothermal deflection spectroscopy (Fig. 2b,c), our findings show consistent variations in these optoelectronic properties that are in good agreement with the XRD results (Fig. 2a and Supplementary Figs. 11 and 12). Notably, formamidinium-based perovskite films processed at an atmospheric temperature of 23 °C exhibit the longest charge-carrier lifetime, the highest PLQY value of 11.6% and the lowest E_U value of 17.6 meV. These noticeable improvements suggest that micron-thick formamidinium-based perovskite films with preferentially oriented high-Miller-index facets can considerably suppress non-radiative recombination because of reductions in defects or imperfections. Further enhancements in optoelectronic properties are observed when these films are passivated with a combination of ethylenediammonium diiodide (EDAI₂) and methylammonium iodide (MAI) (Supplementary Figs. 14–20, Supplementary Table 3 and Supplementary Notes 4–6), and the same preferred orientations for micron-thick perovskite films can be sustained even using this surface passivation (Supplementary Fig. 15b). Moreover, the micron-thick perovskite with preferentially oriented high-Miller-index facets also exhibits an improved electron–hole sum mobility of 36.0 cm² V⁻¹ s⁻¹ when treated with the surface passivation (Supplementary Fig. 21 and Supplementary Note 3).

To explain why micron-thick formamidinium-based perovskite films with preferentially oriented high-Miller-index facets can improve the overall optoelectronic properties, we conducted density functional theory (DFT) calculations and cryogenic transmission electron microscopy (Cryo-TEM) characterizations. First, we calculated the surface energy of formamidinium-based perovskite for the (001)/(002), (110), (111), (210) and (211) surfaces. The allowed chemical potential region for the formamidinium-based perovskite is shown in Supplementary Fig. 22. In our DFT calculations, we considered two different surfaces with Pb-rich and I-rich terminations (denoted as a-term and b-term) and a stoichiometric termination (denoted as st), which is shown in Supplementary Figs. 23 and 24. The atomic arrangements for different surfaces are provided in Supplementary Fig. 24a. Surfaces with the stoichiometric termination on both sides of the slab can be obtained by adjusting the number of I atoms that is self-passivation in nature (Supplementary Fig. 24b,c and Supplementary Table 4).

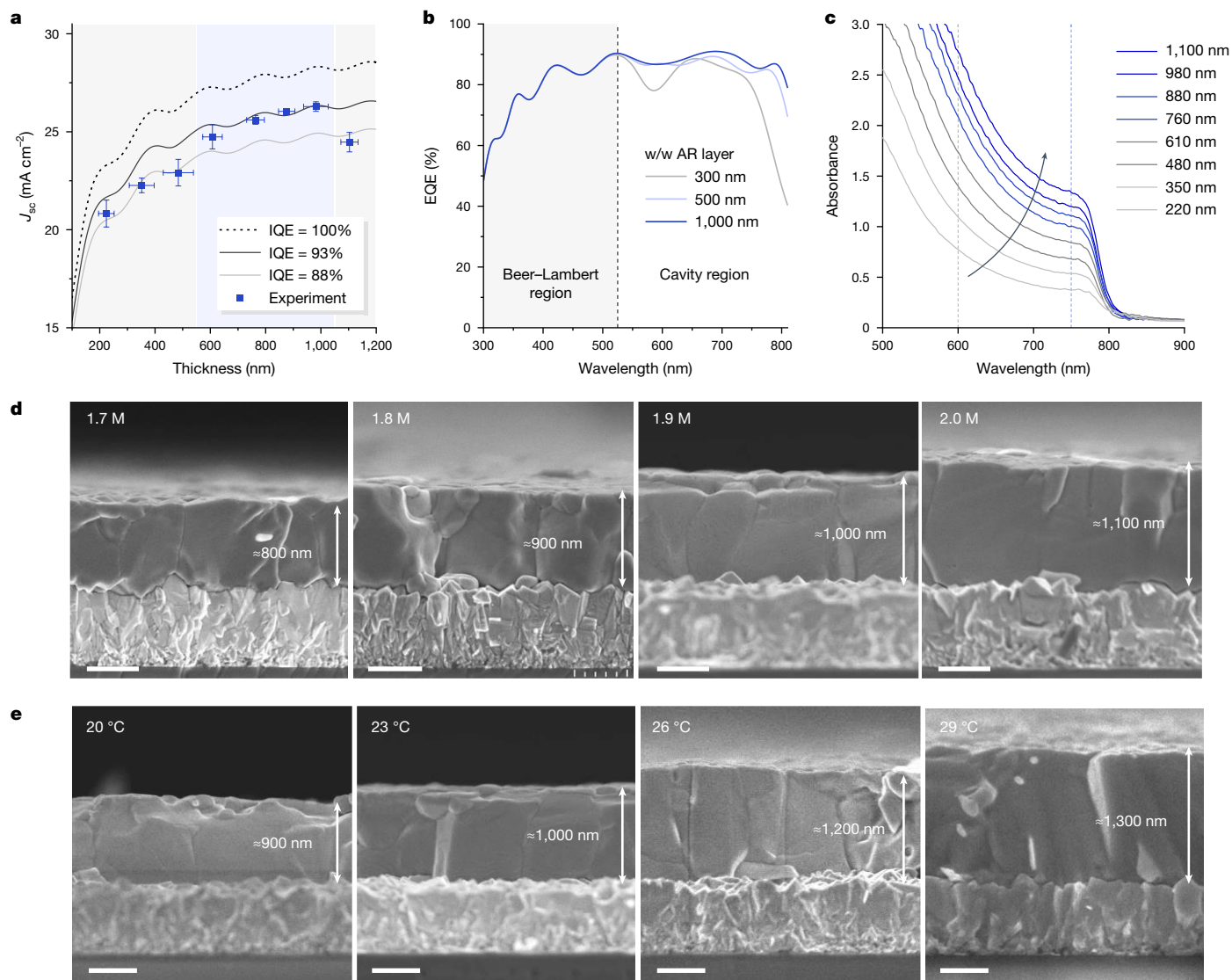


Fig. 1 | Optimal thickness predictions and thickness control. **a**, Comparison of the experimental and calculated J_{sc} as a function of perovskite film thickness. The statistical distributions of the device performance parameters are detailed in Supplementary Fig. 5. The device structure adopted here is $\text{MgF}_2/\text{glass}/\text{FTO}/\text{MeO-2PACz}/\text{formamidinium-based perovskites}/\text{PCBM}/\text{BCP}/\text{Ag}$. PCBM, [6,6]-phenyl-C61-butyric acid methyl ester; BCP, bathocuproine; MeO-2PACz, 2-[2-(3,6-dimethoxy-9H-carbazol-9-yl)ethyl]phosphonic acid; FTO, fluorine-doped tin oxide; MgF_2 , magnesium fluoride; Ag, silver. The optimal MgF_2 thickness depends on the incident light wavelength in Supplementary Fig. 6. MgF_2 (120 nm) that is deposited on the FTO-coated glass with a texture structure is used as an anti-reflection optical coating. The error bars indicate the standard deviation. **b**, Calculated EQE spectra of devices with different

perovskite absorber thicknesses. AR, anti-reflection. **c**, UV-vis absorbance spectra of the thickness-dependent formamidinium-based perovskite absorbers fabricated on $\text{FTO}/\text{MeO-2PACz}$ substrates by using precursor concentrations of 0.9–2.0 M. The composition for formamidinium-based perovskite films is $\text{Rb}_{0.05}\text{Cs}_{0.05}\text{MA}_{0.05}\text{FA}_{0.85}\text{Pb}(\text{I}_{0.95}\text{Br}_{0.05})_3$ (for details, see Methods). The dependence of the solution concentration on thickness is listed in Supplementary Table 2. **d**, Cross-sectional SEM images of 4- μm -thick perovskite films prepared by varying the perovskite concentrations in a solution. Scale bar, 500 nm. **e**, Cross-sectional SEM images of the micron-thick formamidinium-based perovskite processed with a 1.9-M perovskite solution at different atmospheric temperatures. Scale bar, 500 nm.

From the DFT calculation results in Fig. 2d, we find that both the (110) and (211) surfaces have shown abilities to spontaneously adjust the number of I atoms on both sides of the slab, enabling self-passivation in nature. On self-passivation, these (110) and (211) surfaces show a lower surface energy compared with their pristine surfaces. The surface energy of the stoichiometric (211)-st surface is even lower than that of low-Miller-index facets. Furthermore, our DFT calculations show that the surface energy of the (110)-st and (211)-st surfaces is less dependent on chemical potential compared with others (Fig. 2e). This characteristic indicates that the (110)-st and (211)-st surfaces, with their lowered surface energy, can withstand complex environments, thereby positively prolonging the inherent optoelectronic properties.

Moreover, our DFT calculations indicate that the self-passivation on the (110)-st and (211)-st surfaces saturate and merge the unpaired in-gap states stemming from surface dangling bonds into the conduction band and make them more dispersive (Fig. 2f,g), benefiting suppressed non-radiative recombination.

Coherent grain boundaries by Cryo-TEM

Atomic-scale high-resolution Cryo-TEM is a powerful tool for characterizing the fine structural details of electron-sensitive perovskite materials³⁰, covering multiple information from bulk grains to relevant grain boundaries. The high-resolution Cryo-TEM image in

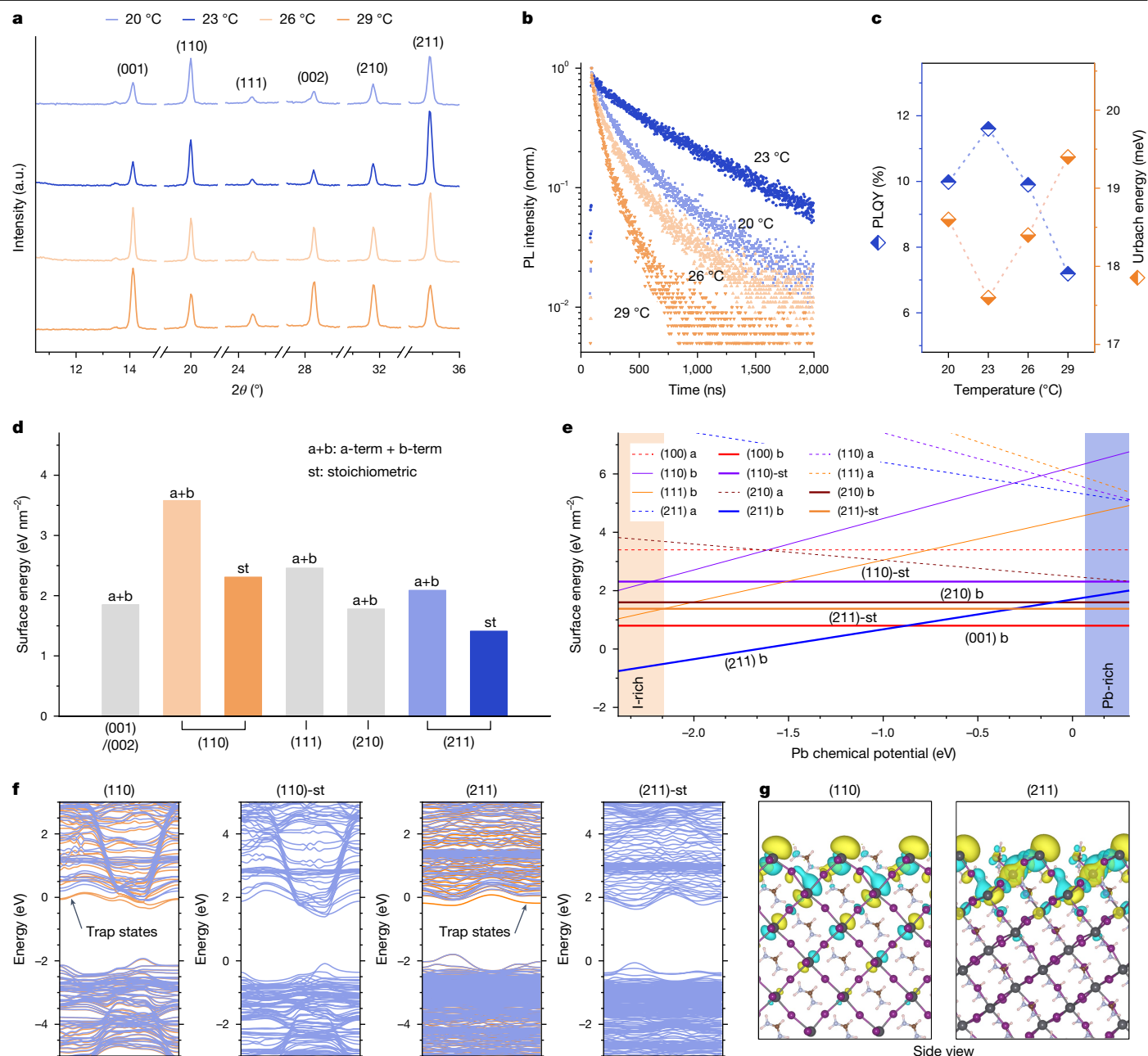


Fig. 2 | Engineering high-Miller-index facets in micron-thick perovskites, alongside DFT calculations. a, XRD patterns of the micron-thick perovskite films processed at different atmospheric temperatures (20 °C, 23 °C, 26 °C and 29 °C). The diffraction peaks are magnified for comparison. The full XRD spectra are shown in Supplementary Fig. 10. **b**, TRPL decays for formamidinium-based micron-thick perovskite films processed at different atmospheric temperatures. PL, photoluminescence; norm., normalized. **c**, PLQYs and Urbach energy (E_U) for formamidinium-based micron-thick perovskite films processed at different atmospheric temperatures. Photothermal deflection spectra used to extract the E_U values are shown in Supplementary Fig. 13. **d**, The calculated perovskite surface energy for the stoichiometric surface structures of the (001)/(002), (110), (111), (210) and (211) orientations. The atomic slab for the ‘a+b’ structures was constructed to contain both the complementary Pb-rich and I-rich terminations. **e**, Surface energy of different orientations and

terminations at the chemical potential condition. **f**, Spin-resolved band structure of the (110) and (211) surfaces for non-stoichiometric and stoichiometric terminations (the two spin components are coloured blue and orange respectively, and the Fermi energy is set to zero). The estimated bandgap in our calculations using the bulk structure (rigorously periodic) is 1.55 eV (Supplementary Fig. 25). The unpaired states arise from the dangling bonds on the surface, leading to trap states for charge carriers. The unpaired surface states of the two surfaces could be saturated on self-passivation to sustain chemical stoichiometry, pushing the surface states to deeper energy levels and making them more dispersive. **g**, Wave function isosurfaces of the unpaired surface states of the (110) and (211) surfaces with non-stoichiometric terminations. Band structures and wave functions of surface states (trap states) of the (001)/(002), (111) and (210) surfaces are shown in Supplementary Fig. 26.

Fig. 3a shows coherent grain boundaries formed between (211)-oriented grains and (001)-oriented grains because of their well-aligned crystal lattice parameters at the interface (Supplementary Fig. 27). The coherent grain boundary is further demonstrated in the atomic-scale

Cryo-TEM image and corresponding fast Fourier transform (FFT) patterns (Fig. 3b,c). Furthermore, DFT calculations show the formation of chemical bonds at the coherent grain boundary between (211)- and (001)-oriented grains (Fig. 3d). Regarding the aggregate

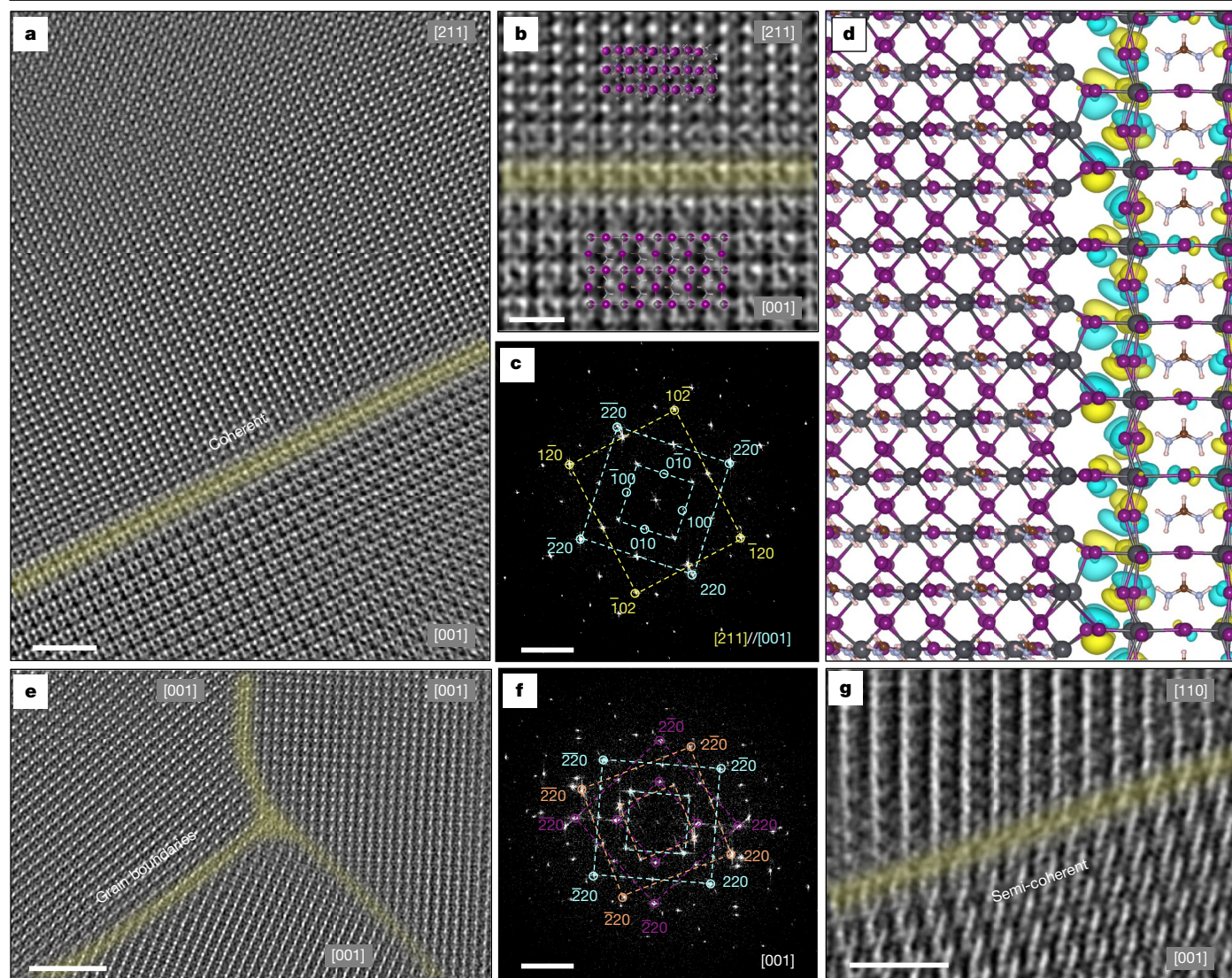


Fig. 3 | Atomic-scale high-resolution Cryo-TEM images and coherent grain boundaries. **a**, High-resolution Cryo-TEM image of the coherent grain boundary between perovskite [211] zone axis and [001] zone axis. The coherent grain boundary is highlighted in light orange. **b**, Magnified view of the coherent grain boundary between perovskite [211] zone axis and [001] zone axis in **a**. **c**, Corresponding FFT patterns of the perovskite [211] zone axis and [001] zone axis coherent grain boundary. **d**, Theoretical atomic model of the coherent grain boundary between perovskite [211] zone axis and [001] zone axis, showing consistency with experimental atomic-scale high-resolution Cryo-TEM results.

e, f, High-resolution Cryo-TEM image (**e**) and the corresponding FFT patterns (**f**) of perovskite incoherent grain boundaries in the [001] zone axis. Three [001] zone axis grains are contacted to form regular incoherent grain boundaries, highlighted in light orange. **g**, High-resolution Cryo-TEM image of the semi-coherent grain boundary between perovskite [110] zone axis and [001] zone axis. The semi-coherent grain boundary is highlighted in light orange. Corresponding FFT patterns are shown in Supplementary Fig. 28. Scale bars, 2 nm (**a, g**); 1 nm (**b**); 2 nm⁻¹ (**c, f**); 5 nm (**e**).

of (001)-oriented grains (Fig. 3e, f), there exist obvious regular incoherent grain boundaries at which dangling bonds, defects and dislocations accumulate, leading to non-radiative recombination and device performance degradation over time³¹. By contrast, when (110)-oriented grains aggregate with (001)-oriented grains (Fig. 3g), a semi-coherent interface without obvious dangling bonds and defects can be observed. As the grain boundaries of the (211)-oriented grains are much more coherent than those of other grain boundaries, the resulting micron-thick high-Miller-index-oriented perovskite absorber would have fewer defects at relevant grain boundaries. Moreover, we have also observed that engineering the high-Miller-index facets in micron-thick formamidinium-based perovskite films, processed in a stabilized atmosphere, can prevent defects and dislocations inside (211)-oriented grains compared with others (Supplementary Figs. 29–31). These improvements in defects, spanning from within

(211)-oriented grains to relevant grain boundaries, can substantially enhance optoelectronic properties.

Device performance and stability

To assess the impact of micron-thick high-Miller-index-oriented perovskites on device performance, we fabricated a series of p-i-n PSCs with a structure of MgF₂/glass/FTO/MeO-2PACz/micron-thick formamidinium-based perovskites (with surface passivation)/PCBM/BCP/Ag (Methods). We measured the current density–voltage (*J*–*V*) characteristics of the devices under a simulated air mass (AM) 1.5 sunlight with 100 mW cm⁻² irradiances and show the statistics of device performances of batch-to-batch devices in Supplementary Fig. 32. The micron-thick high-Miller-index-oriented PSCs, which are processed at 23 °C, present the champion performance among all

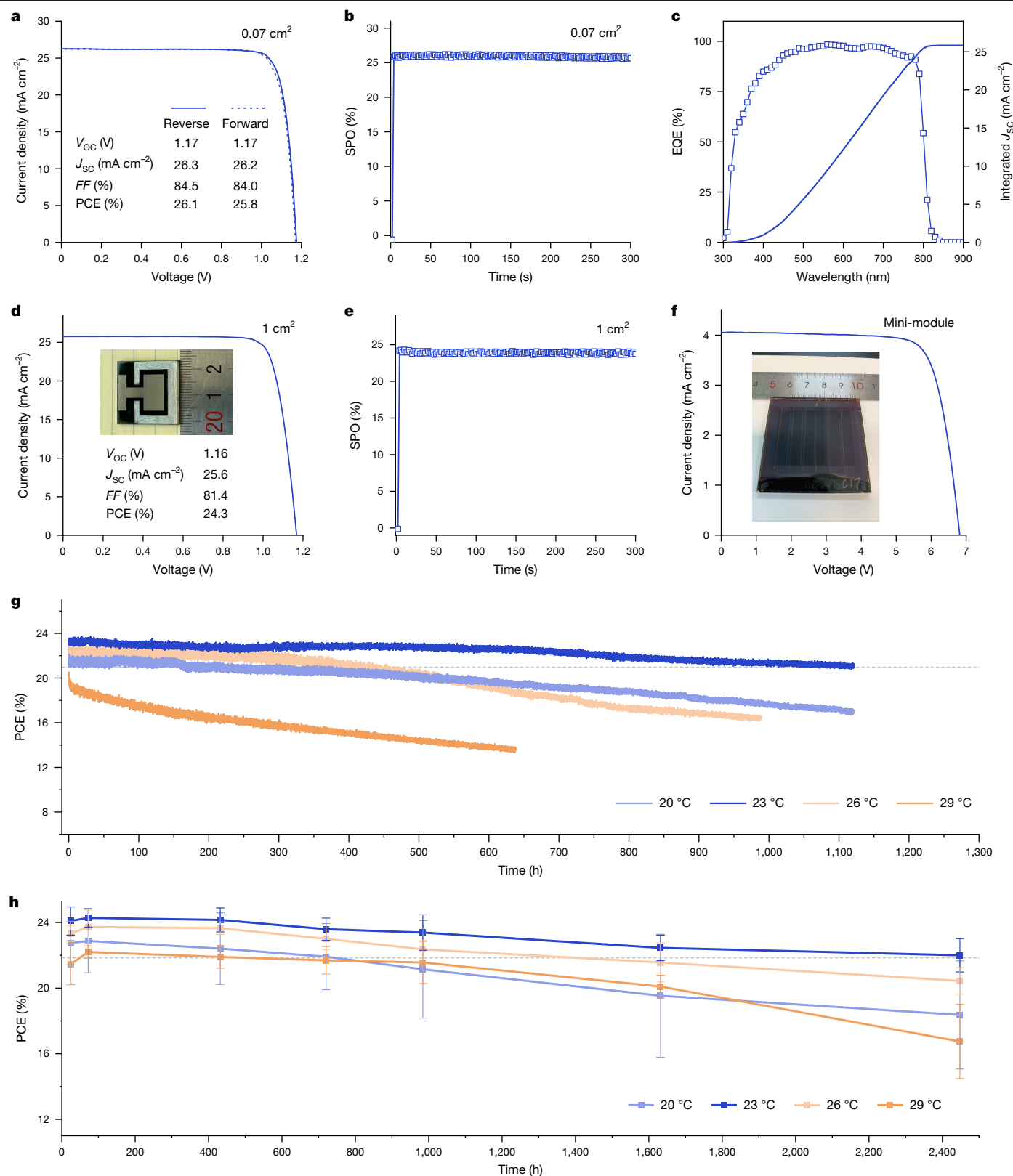


Fig. 4 | Device performance and stability. **a–c**, Current density–voltage (J – V) curves (**a**), SPO (**b**) and EQE spectra (**c**) of the best-performing high-Miller-index-oriented devices. The SPO of 25.9% for the 0.07 cm^2 device is measured at the maximum power point with a period of 300 s. **d,e**, The J – V curve (**d**) and SPO (**e**) of the 1 cm^2 upscaling devices. The SPO of 23.9% is measured at the MPP with a period of 300 s. **f**, The J – V curve of the champion $5 \text{ cm} \times 5 \text{ cm}$ mini-module that has six sub-cells connected in series. **g**, Long-term operational stability of the encapsulated devices at MPP tracking under continuous 1-sun equivalent

illumination by white light-emitting diodes (LEDs) in ambient air at a temperature of $40 \pm 5^\circ\text{C}$ (45% humidity), complied with the ISOS-L-1 protocol. **h**, Thermal stability accelerated ageing test of the unencapsulated devices following the ISOS-D-21 protocol. The evolution of PCEs of the unencapsulated devices under 85°C thermal stresses in the nitrogen atmosphere. Twenty devices for each temperature were used for statistics. The straight dotted line in **g** and **h** represents 90% of the initial PCE for the 23°C devices. The error bars in **h** indicate the standard deviation.

the devices. The change in PCEs is consistent with the case of the V_{OC} values that we determined from the J - V scanned results. By contrast, although the micron-thick perovskite with low-Miller-index orientations processed at 26 °C is much thicker than that of the case of 23 °C, both devices exhibit a comparable average J_{SC} of about 26 mA cm⁻². This could be because of the presence of a small number of ridges (about 5–10 μm in a width) on the surface of the micron-thick perovskite film when processed at equal to or above 26 °C (Supplementary Fig. 33). Figure 4a shows J - V scanned curves of one of the small-area best-performing p-i-n PSCs, delivering the hysteresis-free characteristics once scanned from both forward and reverse directions. The impressive J_{SC} , alongside the high V_{OC} and FF, resulted in a champion PCE of 26.1% (reverse)/25.8% (forward), with the steady-state power output (SPO) of 25.9% (Fig. 4b). We sent one of the champion PSCs with micron-thick high-Miller-index-oriented perovskites to an independent laboratory for certification, in which a certified PCE of 25.85% (a V_{OC} of 1.165 V, a J_{SC} of 26.27 mA cm⁻² and an FF of 84.45%) is confirmed in Supplementary Fig. 34. The certified J_{SC} of 26.27 mA cm⁻² agree well with the integrated J_{SC} by the EQE spectrum over the solar spectrum (Fig. 4c and Supplementary Fig. 35). The certified PCE and J_{SC} values are among the highest values reported to date for p-i-n PSCs.

The external quantum efficiency of electroluminescence (EQE_{EL}), when driven under forward bias current density equivalent to the J_{SC} , is of about 1% (Supplementary Fig. 36). This validates that the corresponding V_{OC} is around 1.16 V. The estimated V_{OC} by measuring the EQE_{EL} is in good agreement with the J - V scanned V_{OC} value. The estimated V_{OC} deficit is as low as 0.38 V, for which the bandgap of the perovskite absorber layer is quantified to be 1.54 eV (Supplementary Figs. 15d and 35). Furthermore, the operando photoluminescence measurements of the full PSCs under 1-sun equivalent illumination (Supplementary Fig. 37) were used to estimate the real-time loss by quantifying the quasi-Fermi level splitting (QFLS) over a range of voltages³². Notably, no substantial QFLS differences for the micron-thick high-Miller-index-oriented PSCs with and without surface passivation have been observed when the bias voltages are less than the voltage at maximum power point (MPP), whereas the noticeable enhancement occurred around the V_{OC} point (Supplementary Fig. 38). Moreover, we observed that the values of QFLS_{OC}- qV_{OC} , representing an additional non-radiative recombination loss originating from either an energetic offset or a lack of charge selectivity at one of the contacts^{32–34}, are smaller in the micron-thick high-Miller-index-oriented PSCs with surface passivation. This indicates marked improvements in band alignment and reductions in non-radiative recombination occur when the micron-thick high-Miller-index-oriented perovskites are treated with surface passivation.

To further demonstrate the commercial potential of micron-thick high-Miller-index-oriented perovskites, we fabricated 1-cm² upscaling PSCs and 5 cm × 5 cm mini-modules. As shown in Fig. 4d, 1-cm² upscaling PSCs (with an aperture area of 1.03 cm²) exhibited a champion PCE of 24.3% (V_{OC} = 1.16 V, J_{SC} = 25.6 mA cm⁻² and FF = 81.4%), corresponding to an SPO of 23.9% (Fig. 4e). Our work also enabled the fabrication of 5 cm × 5 cm mini-modules with an aperture area of 7.20 cm² in a stabilized atmosphere of 23 °C and obtained an efficiency of 21.4% (V_{OC} of 6.82 V, J_{SC} of 29.2 mA and FF of 77.4%) in Fig. 4f. These encouraging results from both large-area upscaling PSCs and mini-modules convince that engineering micron-thick formamidinium-based perovskites with oriented high-Miller-index facets takes a step forward in advancing p-i-n PSCs development.

We further examined the long-term stability of the devices with the micron-thick formamidinium-based perovskites under conditions of continuous illumination or heating. We conducted long-term operational reliability and accelerated stability tests following the ISOS consensus protocols³⁵. For the long-term stability tests, we replaced the Ag electrodes with the more stable chromium (Cr)/gold (Au) bilayer, in which the PCE for the micron-thick high-Miller-index-oriented PSCs

is slightly decreased but remains in the 23–24% range (Supplementary Figs. 39 and 40). We tracked the MPP of encapsulated PSCs under continuous 1-sun equivalent illumination (Supplementary Fig. 41) in ambient air (ISOS-L-1 protocol), with no external heating or cooling. As we show in Fig. 4g and Supplementary Fig. 42, the micron-thick high-Miller-index-oriented PSCs processed at 23 °C show no burn-in degradation in the initial stage and exhibit the most stable operational stability compared with others. Specifically, the micron-thick high-Miller-index-oriented PSCs processed at 23 °C retains more than 90% of the initial PCE (T_{90} lifetime) after continuous 1-sun equivalent illumination stressing for 1,142 h, exhibiting the longest T_{90} among the devices processed at different atmospheric temperatures [T_{90} (20 °C): 749 h, T_{90} (26 °C): 525 h and T_{90} (29 °C): less than 200 h]. Furthermore, we tested a series of unencapsulated PSCs aged at 85 °C in the dark in a nitrogen atmosphere and measured the J - V curves periodically for the thermal stability ageing test (ISOS-D-2I protocol). We tested 20 devices and summarized the results in Fig. 4h and Supplementary Fig. 43. The average PCEs and their corresponding standard deviations show that the micron-thick high-Miller-index-oriented PSCs processed at 23 °C retain, on average, more than 90% of their initial PCEs after 2,448 h. By contrast, the PSCs processed at other atmospheric temperatures retain less than 90% of their initial PCEs after 2,448 h. By tracking the evolution of the key device parameters during the thermal stability ageing test, the drop in device PCEs on thermal ageing at 85 °C and in a dark nitrogen atmosphere results mainly from FF and V_{OC} degradation, without apparent changes to J_{SC} . The micron-thick high-Miller-index-oriented PSCs with the Ag electrode also exhibited decent thermal stability although they show slightly lower device stability (Supplementary Fig. 44). Furthermore, our innovative work can advance the device performance reproducibility when PSCs are processed across all four seasons of a year (Supplementary Fig. 45).

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-024-08159-5>.

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Methods

Materials

Lead diiodide (PbI_2 , 99%), lead dibromide (PbBr_2 , 99%) and 2-[2-(3,6-dimethoxy-9H-carbazol-9-yl)ethyl]phosphonic acid (MeO-2PACz) were purchased from Tokyo Chemical Industry. Formamidinium iodide (FAI), methylammonium iodide (MAI) and methylammonium bromide (MABr) were purchased from Greatcell Solar. Rubidium iodide (RbI, >99.99%) and magnesium fluoride (MgF_2 , 99.99%) were purchased from Alfa Aesar. Caesium iodide (CsI, >99.99%) was purchased from Sigma Aldrich. Ethylenediammonium diiodide (EDAI_2) was purchased from Xi'an Polymer Light Technology. [6,6]-phenyl- C_{60} -butyric acid methyl ester (PCBM) was purchased from Nano-C Tech. Bathocuproine (BCP) and buckminsterfullerene (C_{60}) were purchased from Advanced Election Technology. Ultra-dry dimethylformamide (DMF), ultra-dry dimethyl sulfoxide (DMSO), ultra-dry isopropanol (IPA), ultra-dry ethanol and ultra-dry chlorobenzene (CB) were purchased from commercial sources (Acros). Silver (Ag), gold (Au), copper (Cu) and chromium (Cr) were purchased from a commercial source with high purity. All reagents were used as received without further purification.

Perovskite precursor solution

The formamidinium-based perovskite was prepared using a chemical composition of $\text{Rb}_{0.05}\text{Cs}_{0.05}\text{MA}_{0.05}\text{FA}_{0.85}\text{Pb}(\text{I}_{0.95}\text{Br}_{0.05})_3$ in a solution. In detail, CsI (19.5 mg), RbI (15.9 mg), MABr (8.4 mg), FAI (221.5 mg), PbI_2 (656.9 mg) and PbBr_2 (27.5 mg) were dissolved in 789.5 ml mixed solvent of DMF:DMSO (v:v = 4:1) to prepare the 1.9 M solution. The concentration (ranging from 0.9 M to 2.0 M) of the resulting solution can be adjusted by varying the amount of the mixed solvent of DMF:DMSO (v:v = 4:1). The resulting precursor solution was stirred at 50 °C for 2 h before use and was not filtered before use.

Device fabrication

Before device fabrication, the textured fluorine-doped tin oxide (FTO) substrates with a transmission of 83.8% and a sheet resistance of 6–10 Ω were ultrasonically cleaned using an aqueous detergent solution, deionized water, acetone and IPA for 20 min. Subsequently, the as-cleaned FTO substrates were treated with UV–ozone for 15 min. The following operations were all carried out in a nitrogen-filled glovebox with H_2O and O_2 concentrations of less than 0.1 ppm (at room temperature). MeO-2PACz powder was dissolved in ultra-dry anhydrous ethanol at 0.3–0.5 mg ml^{-1} and then dispersed ultrasonically for 15 min (at 30–40 °C) before use. MeO-2PACz solution (100 μl) was added onto the textured FTO substrates, rested for 15 s, subjected to spin-coating at 4,000 rpm for 30 s and annealed at 100 °C for 8–10 min. After heating, the textured FTO/MeO-2PACz substrates were washed dynamically during the same spin-coating programme by dripping 100 μl of ethanol or IPA twice onto them. Without filtering and dilution, spin-coating of the 0.9–2.0 M perovskite precursor solution was done on the MeO-2PACz-coated textured FTO substrate following an improved processing programme: 1,000 rpm for 10 s with a ramping rate of 1,000 rpm s^{-1} and 3,000 rpm for 30 s with a ramping rate of 3,000 rpm s^{-1} . CB (150 μl) was poured on the spinning substrate at the 30th second during the consecutive two-step spin-coating programme (the temperature of the glove box is around 23 °C). The perovskite film was subsequently annealed at 100 °C for 15–20 min. For the surface passivation, the as-prepared perovskite films were treated with a solution containing 0.5 mg of EDAI_2 , 0.5 mg of MAI, 5 μl of DMF and 1 ml of IPA. The passivation was conducted by depositing the mixed solution (50 μl) onto the perovskite film surface when spun at 4,000 rpm. The film was then annealed at 100 °C for 5 min. Furthermore, the PCBM solution (20 mg ml^{-1} in CB) was subjected to spin-coating on the perovskite layer at 1,000 rpm for 30 s (with a ramping rate of 200 rpm s^{-1}). After that, spin-coating of BCP (1 mg ml^{-1} in ethanol) was done at 5,000 rpm with a ramping rate of 2,000 rpm s^{-1}

for 30 s. A 120-nm thick Ag electrode was thermally evaporated onto the samples in the vacuum chamber ($<4 \times 10^{-4}$ Pa) with a deposition rate of 1–1.2 $\text{\AA} \text{s}^{-1}$. After finishing the device fabrication, MgF_2 anti-reflection coatings were made on the glass side of the textured FTO substrate by thermal evaporation. Note that Cr/Au electrodes were used for the stability assessment to replace the conventional Ag electrodes. The Cr (3 nm)/Au (80 nm) was thermally evaporated onto the samples in the vacuum chamber ($<4 \times 10^{-4}$ Pa) with a deposition rate of 0.2 $\text{\AA} \text{s}^{-1}$ and 0.6 $\text{\AA} \text{s}^{-1}$, respectively. All procedures were performed in the nitrogen-filled glovebox.

For the fabrication of 1 cm^2 devices, solution volumes of the MeO-2PACz, PCBM, BCP and $\text{Rb}_{0.05}\text{Cs}_{0.05}\text{MA}_{0.05}\text{FA}_{0.85}\text{Pb}(\text{I}_{0.95}\text{Br}_{0.05})_3$ were increased properly to fully cover the 1 square inch of the textured FTO substrates. Other processes are the same as the fabrication method of small-area devices. For 5 $\text{cm} \times 5 \text{ cm}$ mini-modules fabrication, the textured FTO substrates (5 $\text{cm} \times 5 \text{ cm}$) were cleaned and treated with UV–ozone in the same way as for the small-area device. Then the MeO-2PACz solution was added onto the pre-cleaned textured FTO substrates and spin-coating was done at 3,000 rpm for 30 s and annealed at 100 °C for 10 min. The formamidinium-based perovskite solution was made using the same chemical composition as the small-area device. The perovskite solution was dissolved in 1 ml mixed solvent of DMF:NMP (v:v = 9:1) and the resulting precursor solution was stirred at 50 °C for 2 h before use. Without filtering and dilution, blade-coating of 15 μl of perovskite precursor solution was done on the MeO-2PACz-coated textured FTO substrate. The gap between the substrate and blade coater was around 30 μm and the coating speed was 8 $\text{mm} \text{s}^{-1}$. This process was done in air with a relative humidity of 50–60%. After the substrate was completely covered with the perovskite precursor, the wetting film was immediately transferred into a vacuum chamber, which was pumped to a gauge pressure of <1 bar in 5 s and kept for 1 min. The perovskite film was subsequently annealed at 100 °C for 10 min in a nitrogen-filled glove box. After cooling down to room temperature, the surface passivation was carried out in the same way as for the small-area device. Afterwards, C_{60} (30 nm)/BCP (5 nm)/Cu (80 nm) was thermally evaporated onto the samples in the vacuum chamber ($<2 \times 10^{-4}$ Pa).

Stability assessments

Thermal stability assessments for unencapsulated devices were conducted by storing the encapsulated PSCs on a hot plate at 85 °C in a nitrogen atmosphere. Photovoltaic performances were regularly measured when cooled down to room temperature. The long-term operational stability of the encapsulated devices was further assessed by MPP tracking under white LED illumination (continuous 1-sun equivalent illumination) in ambient air (relative humidity around 45%), and the ambient temperature in the chamber was around 40 °C during device operation by electrical fans. The encapsulation of the device was done by capping the device with a glass slide, using UV adhesive as the sealant. The encapsulation process was finished in a nitrogen-filled glovebox (with H_2O and O_2 concentrations of <0.1 ppm).

Photovoltaic performance characterization

The current density–voltage (J – V) characteristics were measured with a Keithley 2400 Source Meter under air mass (AM) 1.5G illumination using a Xenon-lamp solar simulator (XES-40S1, SAN-EI). Before the measurement, the light intensity of 100 $\text{mW} \text{cm}^{-2}$ was calibrated using a standard monocrystalline silicon solar cell with a KG-5 filter. The active area of the device was defined with an aperture area of 0.07221 cm^2 . The measurement conditions were as follows: forward scan (–0.02 V to 1.22 V, scan rate 40 $\text{mV} \text{s}^{-1}$ and no delay time) and reverse scan (1.22 V to –0.02 V, scan rate 40 $\text{mV} \text{s}^{-1}$ and no delay time). The external quantum efficiency (EQE) spectra were measured by a QE-R3011 measurement system (Enli Technology). The steady-state power outputs of the non-encapsulated p-i-n PSC devices were measured at the

maximum power point under AM1.5G solar illumination (100 mW cm⁻²) in ambient air, in which the devices were cooled with the wind. The electroluminescence external quantum efficiency (EQE_{EL}) spectra were carried out at room temperature (25 °C) in a nitrogen-filled glovebox. A Keithley 2400 Source Meter and an integration sphere (CME-LP38) coupled with a spectrometer (Flame, Ocean Optics) were used for the measurements. The device was placed on top of the integrating globe and only forward light emissions could be collected. The devices were swept from zero to forwarding bias with a 0.1-V step.

Film characterization

The SEM (HITACHI U8010) was used to investigate the top-view and cross-sectional morphologies with an accelerating voltage of 5–10 kV. Light absorption spectra (300–900 nm) of the as-fabricated perovskite films were then recorded with a spectrophotometer (UH4150, Hitachi). The crystallographic structure was characterized by the XRD instrument (D8 Discover, Burker) using 40 kV, 40 mA radiation (Cu K α , $\lambda = 0.15406$ nm). For the SEM and XRD characterizations, the samples were made by depositing perovskite films onto the FTO/MeO-2PACz substrates. The photoluminescence quantum yields (PLQYs) for perovskite layers were measured under 1-sun equivalent excitation according to the previously reported method³⁶ by combining a continuous-wave laser (445 nm), optical fibre, spectrometer and integrating sphere.

Cryogenic transmission electron microscope

Under ambient conditions (including moisture), organic–inorganic hybrid perovskite exhibits high reactivity and is prone to corrosion. Moreover, the organic cations such as methylammonium and formamidinium are extremely unstable under a TEM electron beam, leading to perovskite becoming amorphous because of electron-beam-induced knock-on damage, radiolysis, charging and heating. Consequently, observing these materials is challenging. To obtain reliable and reproducible results, ultralow-dose Cryo-TEM techniques were used in our work.

A series of sample preparation steps is required before conducting Cryo-TEM measurements^{37,38}. We used a Cryo-ultramicrotome (Leica EMFC7), equipped with a Cryo-box and a Cryo-generation unit for precise cryogenic temperature control, to thin the micron-thick formamidinium perovskite films so that we obtain high-quality, electron-transparent Cryo-TEM samples. The sample surface was aligned with the diamond knife before cutting. We first prepared samples of micro-thick formamidinium perovskite on SAM-coated FTO substrates, using the same process as for XRD sample preparation and device fabrication. The as-prepared perovskite sample was rapidly plunged into liquid ethane in an inert-filled glove box and then transferred to liquid nitrogen for storage under the same atmospheric conditions. A thin slice, approximately 50 nm thick and parallel to the sample surface, was cut at a speed of 0.2 mm s⁻¹ under cryogenic conditions within an inert-filled glove box. This approximately 50 nm slice was then transferred onto TEM grids, ensuring that the sample surface remained parallel to the grids, all within an inert-filled glove box. Moreover, to avoid surface reactions and electrode contamination in ambient air, the samples were stored, transported and loaded into the Cryo-TEM autoloader automatic sampling system under liquid nitrogen conditions throughout the process.

Characterizations of our samples were performed using an aberration-corrected FEI Krios G3i microscope equipped with a Gatan Continuum (1069) EELS spectrometer and a Falcon 4 Direct Detection Device. To ensure high-resolution imaging stability, an automatic liquid nitrogen perfusion system maintained the sample chamber and lens barrel temperatures at approximately –192 °C for several days. High-resolution Cryo-TEM images were acquired with an electron dose rate of approximately 2.5 e Å⁻² s⁻¹ for 5 s.

DFT calculation

Our first-principles calculations are based on the DFT theory as implemented in the Vienna Ab Initio Simulation Package (VASP)^{39,40}. The interaction between the valence electrons and ionic cores was described within the projector-augmented wave approach with a plane-wave energy cutoff of 500 eV (ref. 41). Generalized-gradient-approximation-type cost-effective Perdew–Burke–Ernzerhof (PBE) as a correction functional has been used for the exchange-correlation functional in all geometry optimization and self-consistent calculations. Spin polarization was included in calculations for systems with unpaired electrons. The FAPbI₃ slabs were constructed to simulate the crystalline surfaces of the (001)/(002), (110), (111), (210), and (211) crystallographic orientation with at least 15 Å vacuum space to avoid interaction between neighbouring layers. The Brillouin zone was sampled by the Monkhorst–Pack method and the density of the grid varied from structure to structure. In the structural energy minimization, the atomic coordinates are allowed to relax until the forces on all the atoms are less than 0.01 eV Å⁻¹. The energy tolerance is 10⁻⁶ eV. The surface energy for surface structures with different orientations and stoichiometries was calculated by^{42,43}

$$\sigma(\alpha, \beta, \Delta\mu_{\text{Pb}}, \Delta\mu_{\text{I}}) = \frac{1}{2A} \left\{ E_{\text{slab}}^{\text{relaxed}} [\text{Pb}_{\alpha}\text{I}_{\beta}(\text{FAPbI}_3)_\gamma] - \alpha\mu_{\text{Pb}}^{\text{bulk}} - \frac{\beta}{2}\mu_{\text{I}_2}^{\text{gas}} - \gamma\mu_{\text{FAPbI}_3}^{\text{bulk}} \right\} - (\alpha\Delta\mu_{\text{Pb}} - \beta\Delta\mu_{\text{I}}) \quad (1)$$

with respect to stable phase of components, where $E_{\text{slab}}^{\text{relaxed}}$ is the total energy of the relaxed slabs and A is the surface area, α and β are the excess number of Pb and I that deviate from the stoichiometry of FAPbI₃, $\Delta\mu_{\text{Pb}}$ and $\Delta\mu_{\text{I}}$ are the external chemical potential of Pb and I (Supplementary Note 7).

To visualize and compare directly the stability of different surface structures, we have fixed the chemical potential of Pb and I to the boundary of $\Delta\mu_{\text{Pb}} + 2\Delta\mu_{\text{I}} = -2.05$ eV, the surface energy of the structure as a function of Pb chemical potential is given by

$$\sigma(\alpha, \beta, \Delta\mu_{\text{Pb}}) = \frac{1}{2A} \left\{ E_{\text{slab}}^{\text{relaxed}} [\text{Pb}_{\alpha}\text{I}_{\beta}(\text{FAPbI}_3)_\gamma] - \alpha\mu_{\text{Pb}}^{\text{bulk}} - \frac{\beta}{2}\mu_{\text{I}_2}^{\text{gas}} - \gamma\mu_{\text{FAPbI}_3}^{\text{bulk}} - \frac{\beta}{2}\Delta H[\text{PbI}_2] \right\} - \left(\alpha - \frac{\beta}{2} \right) \Delta\mu_{\text{Pb}} \quad (2)$$

For the structure with a stoichiometric composition, in our case for those with saturated surfaces and the slab of $a+b$, $\alpha = \beta = 0$, the surface energy can be simplified to be

$$\sigma = \frac{1}{2A} \left\{ E_{\text{slab}}^{\text{relaxed}} [(\text{FAPbI}_3)_\gamma] - \gamma\mu_{\text{FAPbI}_3}^{\text{bulk}} \right\} \quad (3)$$

which is calculated with respect to the bulk phase energy of FAPbI₃ and does not explicitly show dependency on external chemical potentials of components.

TRPL decays

The TRPL decays of the thin films on quartz substrates were measured using time-correlated single-photon counting (TCSPC) following excitation by a 398-nm picosecond pulsed diode laser at a repetition rate of 500 kHz and 250 kHz (PicoHarp, LDH-D-C-405M). The resultant photoluminescence signal was collected and coupled to a grating spectrometer (Princeton Instruments, SP-2558), which directed the spectrally dispersed photoluminescence onto a photon-counting detector (PDM series from MPD), in which the time was controlled with a PicoHarp300 TCSPC event timer. The photoluminescence transients were measured at 798, 796.5, 801 and 798 nm, corresponding to the

peak intensity in the photoluminescence spectra of the control and passivated samples, respectively.

Photothermal deflection spectroscopy

The samples were mounted inside a quartz cuvette filled with a thermo-optic liquid (3M Fluorinert FC-72). The excitation source consisted of a halogen lamp coupled to a 250-mm focal length grating monochromator, providing tunable light beam wavelengths for spectral scans, additionally modulated at 13 Hz with a mechanical chopper. The PDS experiments were performed in the transverse configuration, with a probe laser beam (670 nm) passed close and parallel to the sample surface in the area excited with the pump beam. The probe beam deflection induced by heat transfer was detected with a quadrant silicon photodiode and measured synchronously using a lock-in amplifier (Stanford Research Systems SR830).

The Urbach energies (E_U) were extracted from the absorption edge at which the absorption (A) is exponentially related to the photon energy through

$$A(E) = A_0 \exp\left(\frac{E - E_g}{E_U}\right)$$

where A_0 is a term having units of absorption coefficient and E_g is the bandgap.

Photoelectron spectroscopy measurements

Ultraviolet photoelectron spectroscopy (UPS), X-ray photoelectron spectroscopy (XPS) and reflection electron energy loss spectroscopy (REELS) measurements were performed using a Thermofisher ESCALAB Xi⁺ system. The apparatus was equipped with a monochromated Al target X-ray (1,486.7 eV) and He UV light (22.12 eV) as the XPS and UPS excitation sources, respectively. The magnetic immersion lens was used to enhance the collection of the photoelectron for XPS measurement. A −10.0 V bias was applied between the samples and the spectrometer for UPS measurements. The pass energy was set as 20.0 eV and 1.0 eV, and the scanning step was set at 0.05 eV per step for XPS high-resolution spectra and 0.02 eV per step for UPS spectra. A 5-mm Ar ion raster was applied to ensure the uniform region for XPS and UPS measurements. XPS and UPS spectra were recorded in the centre of the sputtering spot. The data analysis was conducted by using the Advantage software package. REELS measurements were performed using a 1-keV electron beam with a pass energy of 6 eV. Argon–2,000-atom gas cluster ion beam sputtering was performed using a cluster size of 2,000 at 4 keV energy to obtain the depth profile data.

OPTP spectroscopy

An amplified laser system (Spectra-Physics, MaiTai–Ascend–Spitfire), with a 5-kHz repetition rate, a centre wavelength of 800 nm and a pulse duration of 35 fs were used to generate the THz radiation using a spintronic emitter. The THz probe was then focused onto the sample, overlaid with a 400-nm excitation pump that was generated using a beta barium borate crystal. The THz radiation transmitted through the sample was then detected by free-space electro-optical sampling in a ZnTe (110) crystal of thickness 200 μm. The OPTP spectroscopy allows us to non-invasively probe the dielectric response of a medium and examine the charge carriers that modify the conductivity of the material at THz frequencies after a well-defined time delay after optical excitation. This approach enables us to investigate the local charge-carrier dynamics and quantify the effective mobilities and bimolecular recombination rates of the photogenerated charge carriers.

Transient absorption spectroscopy

All samples were given glass-lid encapsulation. The output of a Ti:sapphire amplifier system (Spectra-Physics Solstice Ace) operating at 1 kHz and generating about 100 fs pulses was used as a

femtosecond laser source. The 400-nm pump pulses were created by sending the 800-nm fundamental beam of the Solstice Ace through a second-harmonic generating beta barium borate crystal of 1-mm thickness (Eksma Optics). The pump was blocked by a chopper wheel rotating at 500 Hz. The broadband white light was provided by the Disco (Leukos Laser, STM-2-UV) and the pump–probe decay was controlled electronically. The white light was split into two identical beams (probe and reference) by a 50/50 beamsplitter. The reference beam passing through the sample did not interact with the pump, which allows for correcting for any shot-to-shot fluctuations in the probe that would otherwise greatly increase the structured noise in the experiments. Based on this arrangement, small signals with $\Delta T/T$ of about 10^{-5} could be measured. The transmitted probe and reference pulses were collected with a silicon dual-line array detector (Hamamatsu S8381-1024Q, spectrograph: Andor Shamrock SR-303i-B) driven and read out by a custom-built board (Stresing Entwicklungsbüro).

Hyperspectral microscopy

Wide-field, hyperspectral microscopy measurements were conducted using a Photon etc. IMA system. Olympus objective lens with a numerical aperture of 0.9, with 100× magnification, was used for all measurements. To correct for aberrations stemming from the optical elements in the system, the apparatus was calibrated using a reference sample to determine the post-processing parameters that will be used to correct the image distortion. Chromatic aberrations were mitigated by calibrating the z position of the sample for every collected wavelength. To minimize oxygen- and humidity-related degradation behaviour, the samples were stored in a nitrogen-filled glovebox immediately before measurement. A 405-nm continuous-wave laser served as the excitation source. The excitation laser was filtered by a high-quality 405-nm Semrock dichroic mirror to separate the photoluminescence and excitation signals. The emitted signal from the sample was incident on a volume Bragg grating, which spectrally split the light onto a silicon Hamamatsu CMOS camera with an array of $2,048 \times 2,048$ pixels.

A two-step calibration process was used for each objective lens to extract the absolute number of photons at each point. First, a calibrated white light lamp was coupled to an integrating sphere, through the objective lens. The measured lamp spectrum at each point was compared with the known spectrum of the lamp, establishing the relative sensitivity of the system both spectrally and spatially. Subsequently, fibre coupling of a 657 nm laser with a known optical power was done and imaged using the microscope. Hyperspectral measuring the fibre output enabled the conversion between the number of counts and photons at this wavelength. By combining this absolute calibration with the relative calibration from the calibrated white light lamp and integrating sphere, it became possible to accurately quantify the number of photons emitted across the spectrum at each point of the sample. This absolute photon count calibration was essential for fitting the photoluminescence spectrum using the Levenberg–Marquardt least squares algorithm to a generalized Planck law equation. Consequently, we could extract the quasi-Fermi-level splitting and bandgap values at each pixel, following the procedures detailed in ref. 44.

Operando photoluminescence measurements

The operando photoluminescence measurement focuses on measurements performed while the device is under operating conditions. It is based on the simultaneous measurement of absolute photoluminescence spectra and a J – V scan under 1-sun equivalent illumination, as shown in Supplementary Fig. 37. The experimental setup consists of a custom-built system coupled with an integrating sphere (AvaSphere-50-REFL, AVANTES), a spectrograph (Kymera 193i, Andor) and a CCD camera (DU420A-BEX2-DD, Andor) for the photoluminescence spectrum collection. A potentiostat (Autolab PGSTAT 204) is used to apply a voltage and record current; a continuous-wave 532-nm laser (PUS-H-LED) is used for photoexcitation. A Labview code is developed

for system control and data collection. The spectrometer wavelength is corrected using a mercury light calibration source (AvaLight-CAL-MINI, AVANTES); the absolute photon flux is calibrated using a halogen light source (AvaLight-HALCAL-ISP50-MINI, AVANTES). The laser excitation intensity is adjusted to 1-sun equivalent by matching the J_{sc} of a device under a 532 nm laser to that under a solar simulator. During the measurement, a mask smaller than the pixel area is used to ensure that all exposure areas are covered with electrodes. Before the measurement, all devices were encapsulated within a sealed sample holder in a nitrogen-filled glove box to avoid air exposure. The operando photoluminescence measurements were conducted at a scan rate of 10 mV s^{-1} after the devices had soaked for 5 min under the laser light while being held at a bias voltage of 0 V.

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Competing interests H.J.S. is a co-founder and the Chief Scientific Officer of Oxford PV, a company commercializing perovskite solar cells. S.D.S. is a co-founder of Swift Solar. All other authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Deying Luo, Bing Han, Qihuang Gong, Samuel D. Stranks, Henry J. Snaith or Rui Zhu.

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